



Performance, Pricing, and Utilization for Lambda GPU Cloud Instances

William Lorenzo, Drake Hashimoto, Albert Hickey, Syed Muhammad Arham Ali

November 20, 2025

Introduction

As a data analyst and machine learning practitioner, my primary interest in this project stems from exploring Lambda's GPU cloud infrastructure — a platform designed to accelerate high-performance AI workloads. My background includes conducting computational experiments in both R and Python, where I've applied TensorFlow (Developed by Google Brain) and PyTorch (developed by Meta's AI Research) to construct and train neural network models for predictive analytics and deep learning.

This hands-on experience has shown us how crucial GPU acceleration is in scaling AI computations efficiently. Lambda's integration of GPU cloud instances offers a real-world framework to understand how computational hardware, pricing, and performance interact to influence accessibility and innovation in artificial intelligence.

Through this project, our team and I intend to examine how hardware specifications, benchmark performance, and pricing dynamics reflect broader trends in the AI ecosystem — particularly how users and organizations perceive and adopt advanced technologies. Understanding these relationships not only deepens our technical insight into GPU-powered computation but also highlights how cloud-based GPU platforms like Lambda are shaping the future of AI research, deployment, and public perception of emerging technologies.

- *William Lorenzo - Lead Data Analyst, AI Engineer @ Interlinked | Statistics, B.S.: Data Science Concentration @ California State University - East Bay*
- <https://www.linkedin.com/in/william-l-9347a7228/>
- <https://github.com/hiiwilliamss>

Objective of the Study

The objective of this particular study is to approximate whether hardware performance metrics such as `benchmark_score`, `vram_gb`, `num_gpus`, and `power_watts` have a measurable relationship with the `hourly_price_usd` of Lambda GPU cloud instances. The intention is to identify if higher performance GPUs command high hourly rental prices.

Description of Data and Variables:

The dataset, `lambda_ai_gpu_instances.csv`, contains approximately 300 observations of GPU instances carrying variables connected to pricing, hardware specifications, and utilization.

Variable Description Table

Variable	Description	Type
<code>gpu_model</code>	Type of GPU instance	Categorical
<code>vram_gb</code>	Video memory capacity (in GB)	Numeric
<code>num_gpus</code>	Number of GPUs per instance	Numeric
<code>benchmark_score</code>	Performance benchmark score	Numeric
<code>power_watts</code>	Power consumption (in watts)	Numeric
<code>hourly_price_usd</code>	Hourly rental price of GPU instance (USD)	Numeric (Dependent Variable)

Exploratory Data Analysis

Upon constructing further inferential analysis, the dataset was thoroughly explored from using descriptive statistics and exploratory data analysis to summarize distributions and relationships among variables.

```
# Loading Packages
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.4.3

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.0    ✓ stringr    1.5.1
✓ ggplot2    4.0.0    ✓ tibble     3.2.1
✓ lubridate  1.9.3    ✓ tidyr      1.3.1
✓ purrr      1.0.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(broom)
library(reshape2)
```

Attaching package: 'reshape2'

The following object is masked from 'package:tidyr':

smiths

```
library(knitr)
```

Warning: package 'knitr' was built under R version 4.4.3

```
# Loading Dataset
df <- read.csv("lambda_ai_gpu_instances.csv")
```

```
# Examining First Few Rows
head(df)
```

	instance_id	gpu_model	vram_gb	num_gpus	region	hourly_price_usd
1	LMB-100	L40S	96	4	US-East	14.30
2	LMB-101	V100	96	8	Australia	10.54
3	LMB-102	RTX 4090	16	1	US-East	4.41
4	LMB-103	V100	96	8	Europe	6.50
5	LMB-104	V100	48	4	Europe	15.46
6	LMB-105	H100	80	4	Europe	14.54

	avg_utilization_	power_watts	uptime_hours	customer_segment	benchmark_score
1	81.6	1442	496	Individual	9970
2	63.0	825	724	Enterprise	7991
3	81.7	725	204	Individual	7067
4	77.0	1168	291	Enterprise	7577
5	89.0	1350	116	Enterprise	7011
6	55.3	1291	1243	Enterprise	8274

	availability_date	reliability_score
1	2025-08-07	99.6
2	2024-01-19	90.1
3	2023-10-15	100.0
4	2023-12-02	96.8
5	2024-09-24	98.3
6	2022-02-18	92.9

```
# Summary Statistics
summary(df)
```

instance_id	gpu_model	vram_gb	num_gpus
Length:300	Length:300	Min. :16.00	Min. :1.00
Class :character	Class :character	1st Qu.:24.00	1st Qu.:1.00

```

Mode :character Mode :character Median :48.00 Median :2.00
      Mean :54.69 Mean :3.51
      3rd Qu.:80.00 3rd Qu.:4.00
      Max. :96.00 Max. :8.00

region hourly_price_usd avg_utilization_ power_watts
Length:300 Min. : 4.100 Min. : 50.20 Min. : 307.0
Class :character 1st Qu.: 7.737 1st Qu.: 61.48 1st Qu.: 610.5
Mode :character Median :12.435 Median : 75.85 Median : 930.5
      Mean :12.087 Mean : 75.27 Mean : 935.0
      3rd Qu.:16.365 3rd Qu.: 87.20 3rd Qu.:1236.0
      Max. :19.970 Max. :100.00 Max. :1596.0

uptime_hours customer_segment benchmark_score availability_date
Min. : 116.0 Length:300 Min. : 6007 Length:300
1st Qu.: 518.2 Class :character 1st Qu.: 7457 Class :character
Median :1021.0 Mode :character Median : 8678 Mode :character
Mean :1011.1 Mean : 8937
3rd Qu.:1504.2 3rd Qu.:10572
Max. :1999.0 Max. :11948

reliability_score
Min. : 90.00
1st Qu.: 92.47
Median : 94.90
Mean : 94.94
3rd Qu.: 97.62
Max. :100.00

```

```

# Dimension
dim(df)

```

```
[1] 300 13
```

```

# Data Types
str(df)

```

```

'data.frame': 300 obs. of 13 variables:
 $ instance_id : chr "LMB-100" "LMB-101" "LMB-102" "LMB-103" ...
 $ gpu_model : chr "L40S" "V100" "RTX 4090" "V100" ...
 $ vram_gb : int 96 96 16 96 48 80 16 80 96 96 ...
 $ num_gpus : int 4 8 1 8 4 4 8 1 2 4 ...
 $ region : chr "US-East" "Australia" "US-East" "Europe" ...
 $ hourly_price_usd : num 14.3 10.54 4.41 6.5 15.46 ...
 $ avg_utilization_ : num 81.6 63 81.7 77 89 55.3 88.1 77.1 98.1 67.1 ...
 $ power_watts : num 1442 825 725 1168 1350 ...
 $ uptime_hours : int 496 724 204 291 116 1243 492 524 228 1253 ...
 $ customer_segment : chr "Individual" "Enterprise" "Individual" "Enterprise" ...
 $ benchmark_score : int 9970 7991 7067 7577 7011 8274 10869 7814 8426 11297 ...
 $ availability_date: chr "2025-08-07" "2024-01-19" "2023-10-15" "2023-12-02" ...
 $ reliability_score: num 99.6 90.1 100 96.8 98.3 92.9 90.1 97.4 98.3 97.4 ...

```

```
# Checking for NA
na_counter <- sum(is.na(df))
cat("NA:", na_counter)
```

NA: 0

```
# Select only numeric columns for correlation
numeric_df <- df %>%
  select_if(is.numeric)

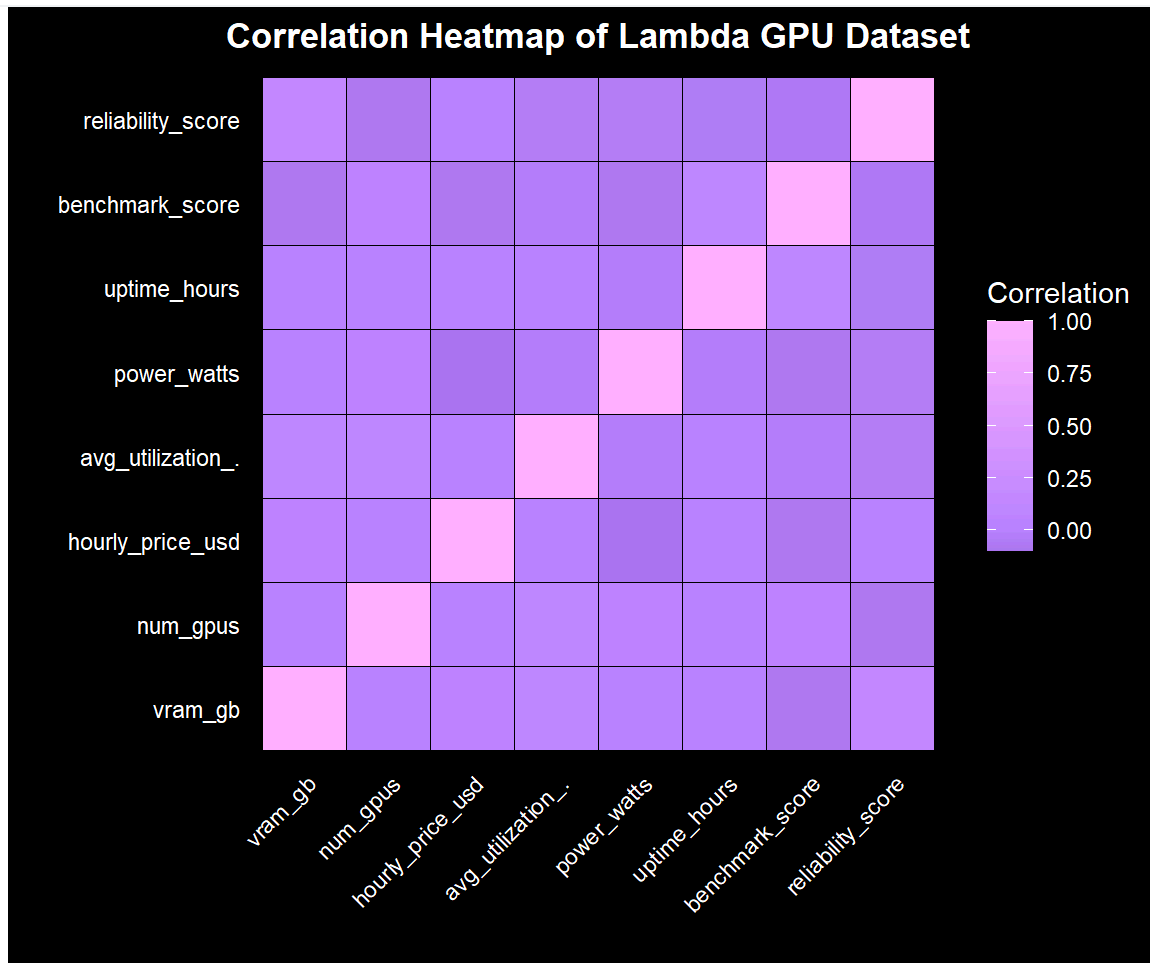
# Compute correlation matrix
corr_matrix <- cor(numeric_df, use = "complete.obs")

# Melt the correlation matrix for ggplot
melted_corr <- melt(corr_matrix)

# Get actual range of correlations for better contrast
rng <- range(melted_corr$value, na.rm = TRUE)

# Plot Heatmap
ggplot(data = melted_corr, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile(color = "black") +
  scale_fill_gradient2(
    low = "#3c096c",    # dark purple for negative
    mid = "#b983ff",    # light purple near 0
    high = "#ffb3ff",   # pinkish-purple for positive
    midpoint = 0,
    limits = rng,       # use actual data range, not -1 to 1
    name = "Correlation"
  ) +
  theme_minimal() +
  theme(
    plot.background = element_rect(fill = "black", color = NA),
    panel.background = element_rect(fill = "black", color = NA),
    panel.grid = element_blank(),
    axis.text.x = element_text(
      angle = 45, vjust = 1, hjust = 1, color = "white"
    ),
    axis.text.y = element_text(color = "white"),
    axis.title = element_text(color = "white"),
    plot.title = element_text(color = "white", face = "bold", hjust = 0.5),
    legend.title = element_text(color = "white"),
    legend.text = element_text(color = "white")
  ) +
  coord_fixed() +
  labs(
    title = "Correlation Heatmap of Lambda GPU Dataset",
    x = "",
```

```
y = ""
)
```



Statistical Inferences

The intention of this particular approach is to carefully examine whether GPU performance variables are crucially associated with hourly price and whether price differences exist between various GPU models.

Correlation and Simple Linear Regression

A Pearson correlation test was performed to assess the linear relationship between `benchmark_score` and `hourly_price_usd`. A simple linear regression was then fit to model hourly pricing as a function of benchmark score.

```
# Pearson Correlation Test
cor_test <- cor.test(df$benchmark_score, df$hourly_price_usd)
cor_test
```

Pearson's product-moment correlation

```
data: df$benchmark_score and df$hourly_price_usd
t = -1.1945, df = 298, p-value = 0.2332
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.18085765  0.04455857
sample estimates:
      cor
-0.06903059
```

```
# Simple Linear Regression Model
lm_model <- lm(hourly_price_usd ~ benchmark_score, data = df)
summary(lm_model)
```

Call:

```
lm(formula = hourly_price_usd ~ benchmark_score, data = df)
```

Residuals:

```
      Min       1Q   Median       3Q      Max
-8.295 -4.405  0.288  4.311  8.383
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   13.7522285   1.4211753   9.677  <2e-16 ***
benchmark_score -0.0001864   0.0001560  -1.195    0.233
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Residual standard error: 4.759 on 298 degrees of freedom

Multiple R-squared: 0.004765, Adjusted R-squared: 0.001426

F-statistic: 1.427 on 1 and 298 DF, p-value: 0.2332

The simple linear regression model is expressed as:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Intercept:

$$\hat{\beta}_0 = 13.7522285$$

- Slope:

$$\hat{\beta}_1 = -0.0001864$$

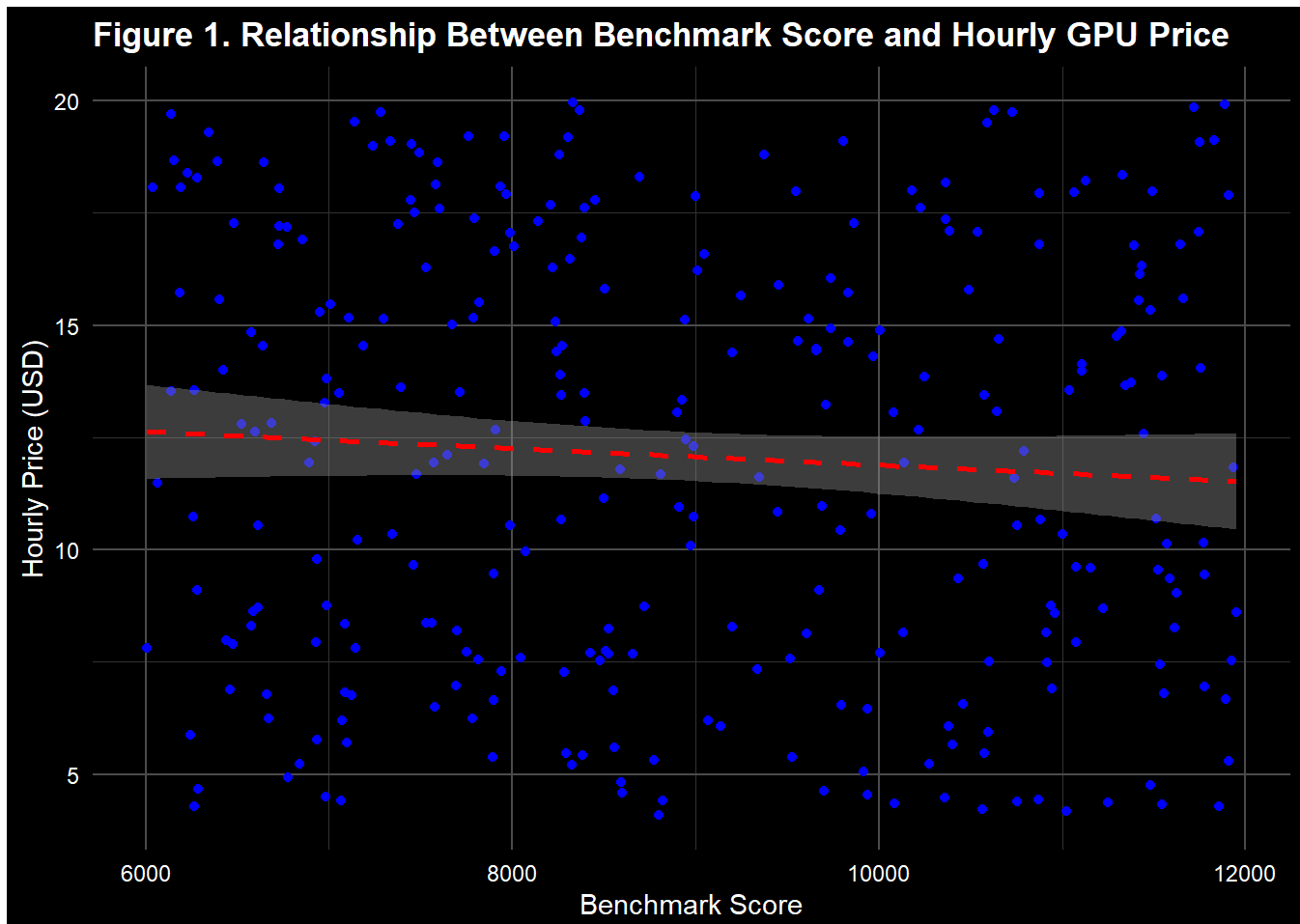
Thus, the estimated regression equation is:

$$\hat{Y} = 13.75 - 0.0001864x.$$

where Y is `hourly_price_usd`, X is the `benchmark_score`, β_0 is the intercept, β_1 is the slope, and ϵ portrays random error.

```
# Linear Regression Visualization
ggplot(data = df, aes(x = benchmark_score, y = hourly_price_usd)) +
  geom_point(col = "blue") +
  geom_smooth(method = "lm", se = T, col = "red", linetype = "dashed") +
  labs(
    "title" = "Figure 1. Relationship Between Benchmark Score and Hourly GPU Price",
    x = "Benchmark Score",
    y = "Hourly Price (USD)"
  ) +
  theme_minimal() +
  theme(
    plot.background = element_rect(fill = "black", color = NA),
    panel.background = element_rect(fill = "black", color = NA),
    panel.grid.major = element_line(color = "gray30"),
    panel.grid.minor = element_line(color = "gray20"),
    axis.text = element_text(color = "white"),
    axis.title = element_text(color = "white"),
    plot.title = element_text(color = "white", face = "bold"),
    legend.position = "none"
  )
```

`geom\_smooth()` using formula = 'y ~ x'



The Pearson correlation has a coefficient of $r = -0.069$ with $p = 0.233$, thoroughly indicating a statistically insignificant negative relationship. The linear regression model produced $\beta_1 = -0.00019$ and $R^2 = 0.0048$, elucidating that benchmark performance explains less than 1% of price variation. Ultimately, there is insufficient evidence to conclude that GPU benchmark performance affect hourly pricing.

Two-Sample t -Test

A Welch two-sample t -test was integrated to determine whether there was a significant difference in mean hourly prices between particular GPU models (A100 and L40S).

The test statistic is defined as:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $\bar{X}_1$ and $\bar{X}_2$ are the sample means, S_1^2 and S_2^2 are sample variances, and n_1 and n_2 are the sample sizes.

substituting values into the Welch test statistic:

$$t = \frac{0.67344}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Using the sample variances and sample sizes computed in R, the resulting test statistic is:

$$t = 0.85174$$

The degrees of freedom (Welch–Satterthwaite approximation) are:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}}$$

$$df \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1-1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2-1}} \approx 139.37$$

Thus, the Welch–Satterthwaite approximation yields

$$df \approx 139.37.$$

For Welch's two-sample t -test, the 95% confidence interval for the difference in population means ($\mu_1 - \mu_2$) is given by

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, df} \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where:

- $\bar{X}_1, \bar{X}_2$ are the sample means
- s_1^2, s_2^2 are the sample variances
- n_1, n_2 are the sample sizes
- $t_{\alpha/2, df}$ is the critical t -value with df degrees of freedom

From the sample:

$$\bar{X}_1 = 12.99290, \quad \bar{X}_2 = 12.31946$$

So the difference in sample means is

$$\bar{X}_1 - \bar{X}_2 = 12.99290 - 12.31946 = 0.67344$$

Using the sample variances, sample sizes, and the Welch–Satterthwaite degrees of freedom $df \approx 139.37$, the estimated standard error of the difference is

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \approx 0.79$$

The critical value for a 95% confidence interval is approximately

$$t_{0.025, 139.37} \approx 1.98$$

Thus, the 95% confidence interval is

$$(\bar{X}_1 - \bar{X}_2) \pm t_{0.025, 139.37} \cdot SE = 0.67344 \pm 1.98 \times 0.79$$

which gives:

$$0.67344 \pm 1.563 \Rightarrow (-0.8898, 2.2367)$$

Therefore, the 95% confidence interval for the difference in mean hourly price (A100 minus L40S) is

$$(-0.8898, 2.2367).$$

Because this interval contains 0, it is consistent with the conclusion that there is no statistically significant difference between the two mean prices.

Under the null hypothesis $H_0 : \mu_1 - \mu_2 = 0$, the test statistic follows a t -distribution with $df \approx 139.37$. The two-sided p-value is then defined as

$$p = 2 \cdot P(T_{df} \geq |t_{\text{obs}}|),$$

where T_{df} is a random variable following a t -distribution with the given degrees of freedom.

Substituting the observed values:

$$p = 2 \cdot P(T_{139.37} \geq |0.85174|).$$

Evaluating this probability (using R or a t -distribution table) gives:

$$P(T_{139.37} \geq 0.85174) \approx 0.1979,$$

so the two-sided p-value is

$$p = 2 \times 0.1979 \approx 0.3958.$$

Thus, the two-sided p-value computed from the t -distribution is

$$p = 0.3958.$$

```
# Welch's Two-Sample t-Test
gpu_subset <- df |>
  filter(gpu_model %in% c("A100", "L40S"))

t.test(hourly_price_usd ~ gpu_model, data = gpu_subset)
```

Welch Two Sample t-test

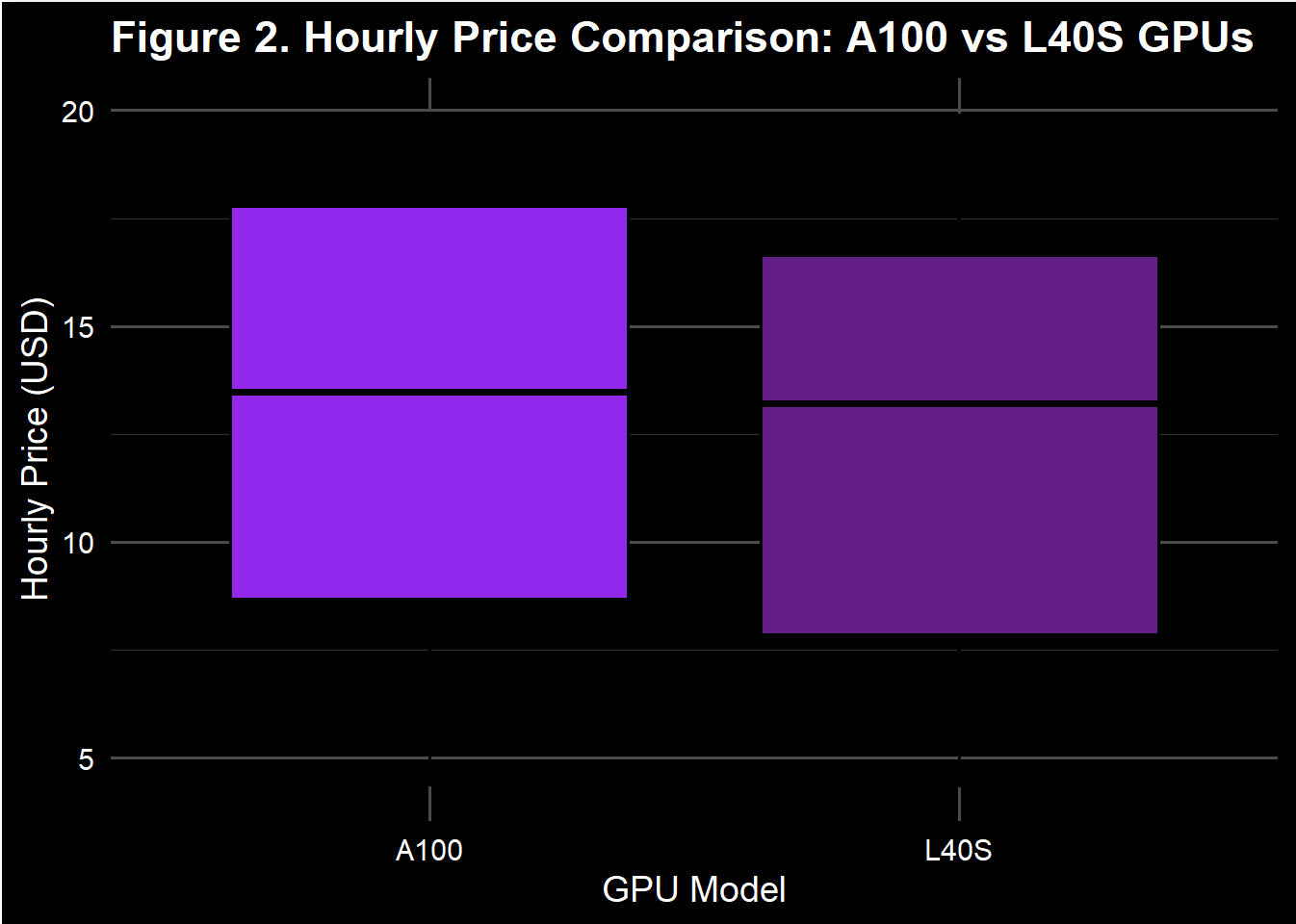
```
data: hourly_price_usd by gpu_model
t = 0.85174, df = 139.37, p-value = 0.3958
alternative hypothesis: true difference in means between group A100 and group L40S is not equal to 0
95 percent confidence interval:
 -0.8897991  2.2366773
sample estimates:
mean in group A100 mean in group L40S
      12.99290      12.31946
```

```
# Boxplot
ggplot(data = gpu_subset, aes(x = gpu_model, y = hourly_price_usd, fill = gpu_model)) +
  geom_boxplot(color = "black") +
  scale_fill_manual(values = c("purple2", "darkorchid4")) + # purple shades
  labs(
    title = "Figure 2. Hourly Price Comparison: A100 vs L40S GPUs",
    x = "GPU Model",
    y = "Hourly Price (USD)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.background = element_rect(fill = "black", color = NA),
    panel.background = element_rect(fill = "black", color = NA),
    panel.grid.major = element_line(color = "gray30"),
```

```

panel.grid.minor = element_line(color = "gray20"),
axis.text = element_text(color = "white"),
axis.title = element_text(color = "white"),
plot.title = element_text(color = "white", face = "bold"),
legend.position = "none"
)

```



```

# Summary Table of Means
mean_table <- gpu_subset %>%
  group_by(gpu_model) %>%
  summarise(
    Mean_Price = mean(hourly_price_usd),
    SD_Price = sd(hourly_price_usd),
    n = n()
  )
kable(mean_table, digits = 2, caption = "Table 1. Mean Hourly Price by GPU Model")

```

Table 1. Mean Hourly Price by GPU Model

gpu_model	Mean_Price	SD_Price	n
A100	12.99	4.81	69
L40S	12.32	4.63	74

$t = 0.85$, $df = 139.37$, and $p\text{ -- value} = 0.396$. There are no significant difference in average hourly price between A100 and L40S GPUs.

The mean hourly price for A100 was \$12.99, while for L40S it was \$12.32. The 95% confidence interval for the difference in means was (-0.89, 2.24), which includes zero, confirming no significant difference.

Conclusion

This analysis indicates that GPU hardware performance metrics—such as benchmark score, capacity of VRAM, number of GPUs, and power consumption—do not significantly predict hourly rental pricing in Lambda’s GPU cloud environment. Both approaches (Pearson’s Correlation and Linear Regression analyses) illustrated no statistically significant relationship between benchmark performance and price, and the t -test confirmed no significant difference between A100 and L40S pricing.

These results suggest that Lambda’s pricing model is not directly performance-driven. On the other hand, crucial variables such as geographic region, reliability, uptime, and market demand likely play a more substantial role in determining hourly rates.